## Behind the scenes

Developers of all the OSs we are talking about are giants. iOS is from Apple, Windows Phone is from Microsoft, Firefox OS is from Mozilla and Ubuntu Touch is from Canonical. Each entity can boast of being highly successful. Yet, none except Apple could stand up to Android in the smartphone industry.

Smartphones are different from desktop computers. People don't buy or assemble a plain smartphone and install the OS themselves. What the manufacturer gives, remains in the device. This is the importance of the alliance behind Android.

Android is not powered by Google alone. The Open Handset Alliance plays a key part in the development of Android. According to Wikipedia, "The Open Handset Alliance (OHA) is a consortium of 84 firms to develop open standards for mobile devices. Member firms include Google, HTC, Sony, Dell, Intel, Motorola, Qualcomm, Texas Instruments, Samsung Electronics, LG Electronics, T-Mobile, Sprint Corporation, NVidia, and Wind River Systems." The article also adds: "Android, the flagship software of the alliance, is based on an open source licence and has competed against mobile platforms from Apple, Microsoft, Nokia (Symbian), HP (formerly Palm), Samsung Electronics/ Intel (Tizen/Bada) and BlackBerry."

It is clear that the backing of handset manufacturers plays a key role in the success of Android.

## Interface

It is really disappointing that no OS developer seems to be introducing something original and innovative in the case of interface design. Lock screens, main menus, notification bars... Every manufacturer seems to be imitating others.

In the case of desktop OSs, you can see differences between each version, distro and desktop interface. But in the case of mobile OSs, one might ask what the differences are. Maybe this is a result of small screen sizes and the limited number of keys available.

However, Android has contributed a lot in making the mobile interface interactive. Newer OSs can't beat it if they don't introduce something revolutionary.


Figure 1: Different OSs seem to follow a common pattern in interfaces

## Language support

Smartphones are not for business executives alone. Even people who are unfamiliar with English use smartphones. Although many of them tend to stick with the English interface, it is still an advantage that a system ships non-English interfaces. Android supports its interface in 70 languages, which means a vast group of people can experience the OS in their mother tongues.

## App stores

The functions of an operating system include executing apps and arranging system resources for them. But in a practical sense, without apps, what is an operating system for?

It is estimated that more than a million apps have been developed for Android. iOS pioneered the app store idea, but now Android has gone much further. Lack of a good app store is the key disadvantage of Windows Phone. When Android users are in need of some tool, they simply go to Play Store and find it there. But that is not the case with many other OSs.

Firefox Marketplace is where you find Firefox apps. It is growing, but is still very small. It's the same situation with Ubuntu Touch Apps.

## Security

Android apps run in a sandbox, an isolated area. But explicit permissions can be granted during installation. This doesn't happen without the permission from the user. But there is no built-in encryption facility for data storage. iOS seems to have some limited security features, but all major OSs require third party support here.

Android malware does exist and Wikipedia says that "…research in 2015 concluded that almost 90 per cent of Android phones in use had known but unpatched security vulnerabilities due to lack of updates and support."

It is clear that Android doesn't offer the security of desktop GNU/Linux distros.

## Hardware

Android supports a large number of platforms including 32- and 64-bit: ARM architectures, x86, x86-64, MIPS and MIPS64. This enables Android to be used in devices other than limited mobile phones. From 2012 onwards, Android tablets using Intel processors have been available. Android-x86 can run on desktop PCs.

Android's RAM requirements start from 512MB (normal cases), which is low compared to Ubuntu Touch (1GB).